BEU CONNECT

Bihar Engineering University, Patna

B.Tech 6th Semester Exam., 2025

Code: 110601 ***Time: 03 Hours***

Semester: VI ***Subject: Measurements and Instrumentation*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) The smallest change in a measured variable to which the instrument responds is
(i) Sensitivity (iii) Resolution
(ii) Drift (iv) Dead-band

(b) Which type of error cannot be eliminated?
(i) Gross (iii) Systematic
(ii) Random (iv) Zero error

(c) Higher C_p value means
(i) Better process capability (iii) Poor capability
(ii) High variation (iv) Low precision

(d) Hall sensor is used for measuring
(i) Temperature (iii) Magnetic field
(ii) Pressure (iv) Flow

(e) Hot wire anemometer is used for
(i) Air velocity (iii) Displacement
(ii) Pressure (iv) Torque

(f) Loading effect is lower in
(i) High resistance voltmeter (iii) Low resistance voltmeter
(ii) High resistance ammeter (iv) None

(g) Moving iron instrument is used for
(i) AC only (iii) DC only
(ii) Both AC and DC (iv) None

(h) LCR meter is used to measure
(i) Voltage (iii) Resistance
(ii) Inductance & Capacitance (iv) All

(i) Which instrument measures peak voltage?
(i) DMM (iii) CRO
(ii) Megger (iv) Clamp meter

(j) Sonography is based on
(i) Ultrasound (iii) X-rays
(ii) Magnetic resonance (iv) CT techniques

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

BEU CONNECT

Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Differentiate between true value, measured value, and error. How do these affect the reliability of instruments?

(b) What is resolution? Explain how resolution affects digital and analog measurement systems.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Discuss static sensitivity, scale readability, and drift with examples from temperature sensors.

(b) What is Six-Sigma? Explain its significance in measurement system analysis.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Explain the calculation of C_p and C_{pk} for a manufacturing process with an illustrative example.

(b) Explain the working principle of RTD, and thermistor for temperature measurement. Compare their characteristics.

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) Describe the construction and working of strain-gauge-based pressure transducers.

(b) Explain the difference between active and passive transducers with suitable examples.

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Describe digital voltmeters (DVM) and digital ammeters (DAM) with block diagrams.

(b) Describe the working principle of Potential Transformers (PTs) and their applications.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Explain the block diagram and working of Digital Storage Oscilloscope (DSO).

(b) Explain errors and limitations in RLC measurement instruments.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) Explain the construction and working of True RMS meters.

(b) Explain the basic components of biomedical instruments with block diagrams.

Q.9 Solve both questions : **[7 x 2 = 14]**

(a) Explain EEG measurement techniques and waveform analysis.

(b) Explain principles of CAT (CT) scan techniques and image reconstruction.

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store